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Demonstrate the Predictive Model of Machine Learning in Smart Grids to Process Large Electricity Data According to Power Demand and Supply

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Abstract

Smart grids are the improved and featured version of traditional power grids associated with the internet of things and machine learning adoption. Forecasting electric consumption using smart meter data is one of the most prevalent applications of machine learning. Detail examination of customers' smart meter data is required to discover essential variables and the source of variation between appliance consumption levels and customer demand. With the ML adoption in smart grids, power load prediction has become possible for a particular client or sector. This paper presented an ML based prediction model of electricity load with time series data taken from the smart meter for understanding the power demand and supply for different consumers. In this regard, two machine learning models LSTM and CNN have been performed for prediction accuracy terms of performance improvement, and it has been found that CNN based model is more accurate than LSTM. The accuracy of the prediction of consumers' electric consumption using CNN is almost 84%, which is significantly higher than previous research in the same field. The model's performance is measured in terms of MSE which is the loss function for the predictive models. The performance of prediction by comparing the predicted data and original data is also demonstrated in this paper.

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Keywords: Smart Grid; Machine learning; Long Short-Term Memory; Convolution Neural Network; Smart Meter; Mean Square Error; Demand Response.

1. Introduction

The Internet of Things (IoT) makes it possible for real-world devices to communicate to each other. Several remote computing applications have been made with the help of this feature. These applications are used to get information from places that are far away and use communication links to do so [1]. This information has been collected on a centralized server. Further by using application or machine learning techniques the collected data has been refined and analysed for recovering the required insights. This technology these days is utilized in several applications for monitoring, surveillance, and other automation and control purposes. One of the great uses of IoT is found in smart power grids. With their new networking features, these smarter versions of electrical grids can help people use electricity more efficiently and pay less for it [2]. With the increasing demand for energy, we must discover more efficient methods of generating, storing, and transporting electricity. The most effective strategy for achieving this goal would be for governments to install smart grids in place of conventional electrical grids.

Smart grids consist of the same transmission lines, transformers, and substations as conventional power grids. Smart Grids include IoT devices that can communicate with one another and with consumers. Smart grids deliver electricity to consumers through digital communication that facilitates the monitoring and evaluation of the electrical supply. IoT must be used in the smart grid for this feature to work. The goal of making smart systems is to save energy and protect the environment. Additionally, it makes it possible to automate management procedures, lessen power outages, and simply incorporate diverse renewable energy sources. Key characteristics of a smart infrastructure include:

Managing loads: A power grid must always be able to give power. Smart grids can tell consumers to change how they use electricity when the grid is busy.

Providing assistance for demand response: It recommends using low-priority devices while rates are low to lower electricity bills. This helps in real-time consumption and charge analysis.

Power Generation Decentralization: It makes it easier to integrate renewable energy sources like solar panels into decentralized power systems.

Components of smart grid and its features are depicted in the given below diagram. The figure 1 illustrates how it is interconnected with other fields and playing a crucial role everywhere, including in the fields of distributed power generation, solar energy, petrol pumps, smart meters, and automation, IOT devices, nuclear energy, factories, thermal power, wind energy, smart houses, and integrated communication systems.

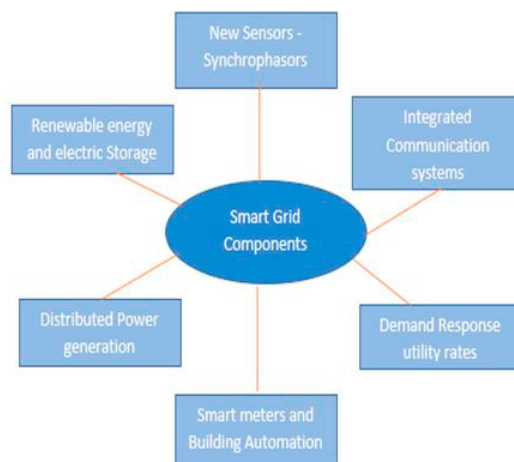


Fig. 1. Smart Grid Features [1]

Smart grids can integrate energy production from multiple sources. This enables greater energy output that can meet the rising demand for electricity. Smart transmission lines are used in smart grids to deliver power within a certain voltage range. This reduces power loss caused by heat in the power lines. With smart meters, consumers can observe their power consumption and integrate into their homes. This allows them to better manage their energy consumption.

The maintenance staff can execute a variety of preventive checks to ensure that the grid remains operational. Additionally, it allows for the storage of excess energy, making it compatible with modern backup power systems [3],[4]. There are three main parts to smart grids:

- Instrumentation and controls
- Devices for Networking and Computing
- Applications of IoT

It measures electricity consumption and supply using measuring and control devices like power sensors. They transmit this data to computing systems, which determine the most effective route to supply the power. The control devices receive this information and adapt as necessary. Finally, the data is delivered to IoT applications that consumers and grid workers utilize via networking tools. This procedure can facilitate functionality based on the needs of the population [5].

In this paper, we improve the accuracy of prediction of load consumed by the user of different sectors according to the requirement using machine learning methods. The hypothesis is to utilize the ML algorithm for sensing and predict the possible future power demands from industry as well as domestic infrastructure. Additionally, maintain the optimal power supply to both the infrastructures. In this context, a proactive technique of resource management has been proposed for design and implementation. The proposed technique utilizes the future workload predicted by the ML techniques and then utilized to increase or decrease power supply according to the requirement of the sectors.

2. Review of literature

In [1], the structure and parts of a smart grid have been addressed. With IoT designs and IoT usage standards and criteria for SG, and smart grid related IoT services and applications. This study also identifies unresolved problems, difficulties, and potential future research topics for IoT-supported SG systems [3], [4]. Reference [5], discussed about the technology that support IoT, like cloud computing and other data analysis tools. Then, problems with IoT deployment in the energy sector with safety and privacy issues were discussed. So, within this context, one remedy for these issues was blockchain technology mentioned. In [6] The authors provide a thorough analysis of the development of SG in relation to the use of IoE systems and the crucial IoE elements for decarbonization. Additionally, mathematical simulation models are offered to assess the contribution of IoE to the reduction of CO₂ emissions. Reference[7] examined at six machine learning (ML) techniques for power prediction in smart grid systems, including support vector machines (SVM), k-nearest neighbour, linear regression, and artificial neural networks. Performance is efficient when Renewable Energy Sources (RES) are integrated. Utilized the multiple knapsack theory to formulate the issue in a way that maximizes the capacity of the electricity consumer. In [8], This study provides a thorough analysis of the load forecasting category, performance indicator computation, data analysis process, load forecasting utilizing traditional meter information, technology utilized to carry out the operation, and obstacles. The significance of smart meter-based load forecasting is then explained, along with the many methods that can be used. In addition, the advantages of smart meter load forecasting over traditional meter load forecasting are discussed in this work. For multivariate time-series forecasting, a canonical and simple long short-term memory (LSTM) DL model for electrical load is created in [9]. The performance of LSTM models for a one-step-ahead prediction is assessed using one available dataset from Europe as a benchmark. A benchmark for hybrid LSTM-optimization methods for multivariate energy time-series forecasting in power systems can be established using the results that have been reported. Reference [10] focuses on the drawbacks of clever DR algorithms. Additionally, a comparative analysis of various intelligent DR methods is given. Quantum methods are advised to reduce the computational load for DR in a future smart grid in light of the conclusions. In [11], An LSTM neural network and multi-factor analysis-based short-term power load forecasting model is suggested. The suggested model is examined, validated, and contrasted with other models using the power load data from the 2016 Electrician's Cup modelling competition. In [12], it has been discussed that SG has a security clause in place when it is being utilized for risk analysis and decision-making. Examine a use case for automated voltage regulation. further discovers the advantages of 5G for security and suitability (DG). In [13], A mixture of the wind-driven optimization (WDO) and bacterial foraging optimization (BFO) algorithms called the wind-driven bacterial foraging algorithm (WBFA) has been used and in reference [14], ML techniques have been presented and discussed to process real-time data and the SG architecture to forecast energy. Explain the connection between the SG, IoT, and ML components. Also, provide an estimate of the energy usage using Recurrent Neural

Network and Long Short-Term Memory. Reference [15], presented the Short-Term Load Forecasting (STLF). And data from electricity and weather conditions have been taken as load. ML methods such as Linear Regression (LR), Support Vector Machines (SVM), Ensemble Bagged (EB), Ensemble Boosted (EBo), Gaussian Process Regression (GPR), and Fine Tree (FT) are implemented. In [16], work has been proposed for developing the infrastructure for the electric utility that focuses on the distinctions between conventional and smart grids, the incorporation of renewable energy sources, and the regulation of electrical usage to conserve resources. Reference [17] shows the policies and regulations about the smart grid and energy storage. The Multidirectional LSTM technique has been suggested in [18] to forecast the network of SG stability. The outcomes demonstrate how well the MLSTM functions. The fundamental architecture of internet of things for power transmission and monitoring have been discussed in [19]. The Machine learning technique have been discussed in [20] for energy management model. Also discussed about service level agreement between prosumers and smart grids which is helpful for both. A model has been designed in which GA is combined with data on load, price, and renewable energy sources. Adopting the seasonal effect of power usage involves seasonal fluctuations. In [21], discussion done on rules and regulations of policy depending on a system for managing energy in which consumers and aggregators participate in a subsidiary program. The purpose is to help people comprehend how to use the patterns, threats, and problems. This paper [22] offers a thorough analysis of the IoT and EC-based smart grid technologies. In the article, it is underlined how emerging technologies are developing, how EC-IoT-based SG works, and what is needed to put it into practice. A machine learning (ML) method have been developed in [23] to identify a cyber deception attack in state estimate and measurement data gathered by a smart grid. To increase detection precision and cut down on complexity, they adopt GA-feature selection. Using common IEEE bus test systems, the scheme is assessed. Techniques to identify bad behaviour of SG devices have been presented in [24] based on convolution, that makes use of the system and library at the operating system kernel level. The framework can recognize malicious devices based on the various attributes. The framework produces extremely accurate results and defeats faulty equipment. A study on the use of ML in intelligent applications have been provided in [25]. The ML methods applied to the attack analysis. They describe the applications that technology can be used in. Future problems and difficulties are then investigated. In [26], a load forecasting model have been implemented for electric appliances based on machine learning techniques. In this smart meter dataset are used to predict the load consumption by different consumers. Two data sets have been used for the load forecasting and MATLAB is used as simulation tool.

3. Proposed work and methodology

The objective of the proposed study is to analyze smart meter data to comprehend the demand and supply patterns of various consumers. Therefore, the exploratory investigation of the data has been conducted initially. we utilize a Plug Readings of Home Area Network (HAN) dataset. That dataset is publicly available at [27]. The dataset is provided by the Australian Government Department of Climate Change, Energy, the Environment, and Water. The dataset was created on 09-09-2015 and last updated on 11-04-2022.

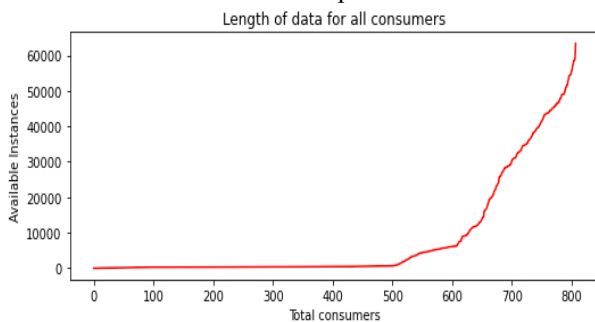


Fig. 2. consumers data instances

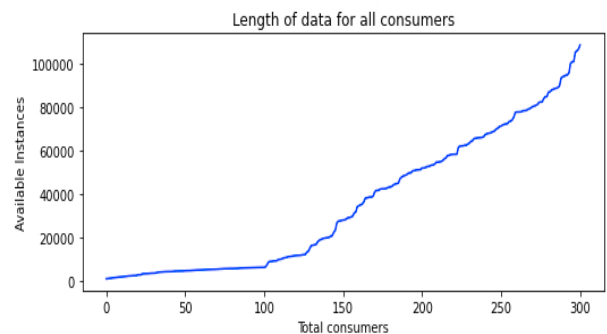


Fig. 3. shows the reduced set of data

The dataset consists of CUSTOMER_ID (Cid), READING_TIME (date), PLUG_NAME (PName), READING_VALUE (Reading), CALENDAR_KEY (CKey) and RECORD_COUNT (Rcount) attributes.

There are 10828120 individual data records throughout the overall dataset. The total number of 808 customers are included in this dataset. On the other hand, some of the customer data is extremely limited and includes less than 500 occurrences or readings. Figure 2 shows the instances or readings of data that are readily available for each consumer. The simulation has been done in python software and according to data set results have been generated and showing in figure 2. Now, we have decided to remove those consumers who has less amount of data; thus, we apply a threshold (1000 instances) for reducing the amount of data. After removal of low informative instances, we again get the simulation graph after execution the data which is shown in figure 3.

Now the dataset contains only 300 consumer's data, now the dataset contains a total of 10639969 instances of data. The given dataset is a time series data thus we set the time stamp as index of the dataset. In order to understand the pattern seasonal pattern of power demand we have plot the dataset according to the time index. The time index-based data is demonstrated in figure 4. In the data plot, we can see there are fewer samples available in data between the dates 07-2012 to 06-2013. Therefore, we reduce these instances of data. After removal of dataset samples, we have found the following seasonal plot of data as given in figure 5. The reduced dataset consists of a total of 10532290 instances.

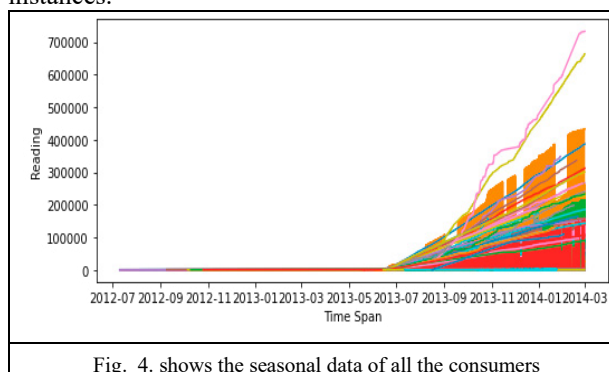


Fig. 4. shows the seasonal data of all the consumers

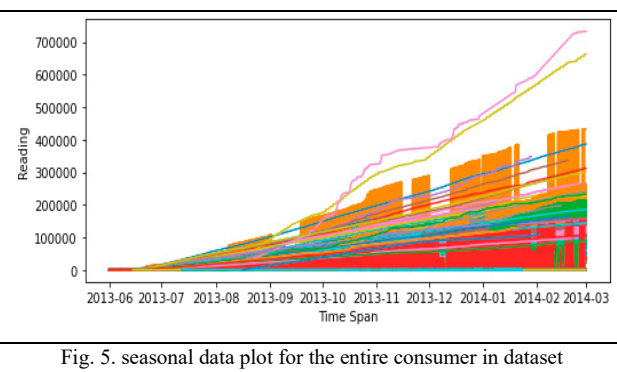


Fig. 5. seasonal data plot for the entire consumer in dataset

Now we are performing the study of client id-based energy demand thus we plot the data of reading (demand) and the clients. The figure 6 demonstrates the data distribution of client id and their relevant demand.

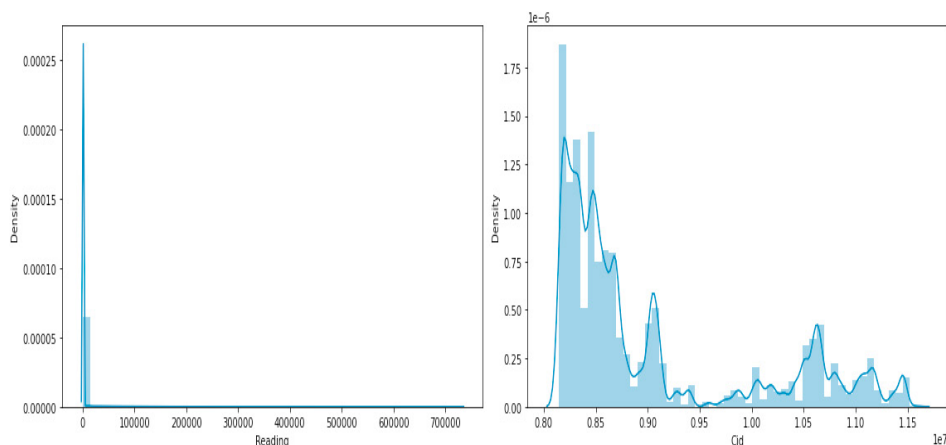


Fig. 6. shows the values distribution of unique client and reading.

According to the values distribution we have found the dataset consist of highly irregular patterns. Therefore, we have considered the method for outlier removal. The dataset therefore regulated using a thresholding scheme after and before outlier detection. After that we have performed the data capping to improve the information of the dataset. After capping of the data, we found a total of 7904325 instances of data. The final dataset with maximum information is demonstrated in figure 7.

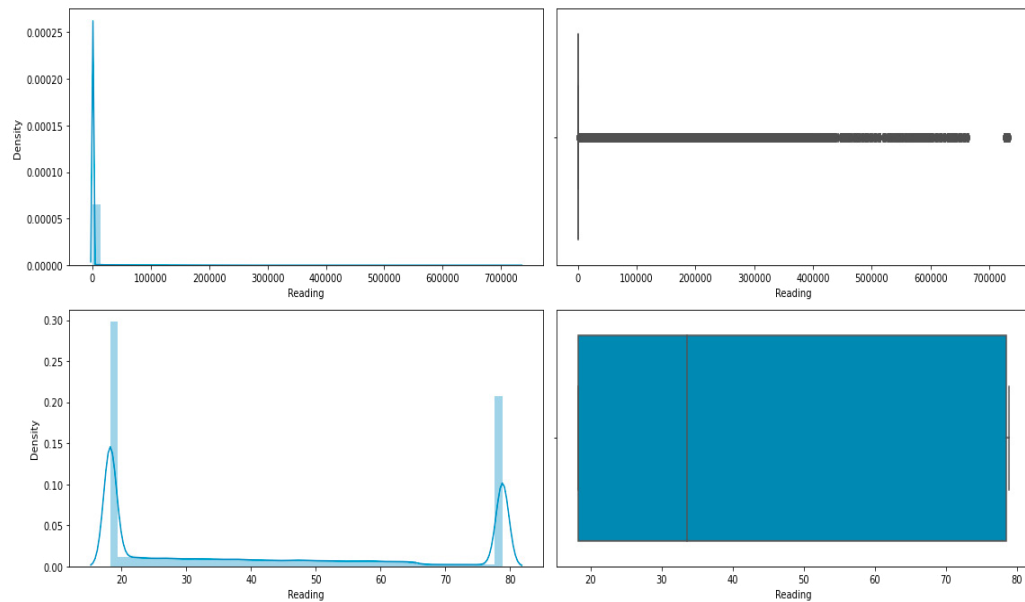


Fig. 7. Showing dataset after capping

Now we have selected an individual client for performing training of the long- short term memory (LSTM) model and Convolutional neural network (CNN). Thus, we have selected the Cid = 8145135 which consist of 47226 samples. After extracting the individual client data, we are transforming the data to train the model and performed training.

3.1. Flow chart of proposed model

The collected data have been pre-processed prior to the application of the methodologies used to improve the performance of the forecasting model. To ensure accurate, efficient, and relevant analysis, datasets may require pre-processing procedures. Before running machine learning algorithms on raw data, a data cleaning and preprocessing step is used. The term "data cleaning" describes procedures for locating, erasing, and restoring missing data. The proposed predictive model for understanding the power demand and supply is demonstrated in figure 8. This proposed model has been simulated in python software. The algorithms have been implemented in google Collaboratory which provides a virtual online environment for programming. We have used machine learning methods for the proposed model. In this model the smart meter dataset is used. The smart meter dataset is a huge dataset which contains the reading of the meters according to the appliances used in smart homes (table1). This data set also contains client-based data. First the data pre-processing is performed for cleaning the dataset which is shown in table 2. After data cleaning, for finding better prediction yield and accuracy the outlier has been identified and removed using a thresholding-based technique. That process improves the significance of the dataset. Finally, the dataset transformed from time series dataset to sequence of data for preparing the training and testing sets. The training set contains 75% of data and 25% of the dataset is used for testing. The training set has been used with the two kinds of machine learning models namely LSTM and CNN based neural network. After training the validation of the model is performed and the results have been computed. The results are calculated in terms of Mean Square Error (MSE). The next section provides the comparative performance of the models for both training and testing of the model.

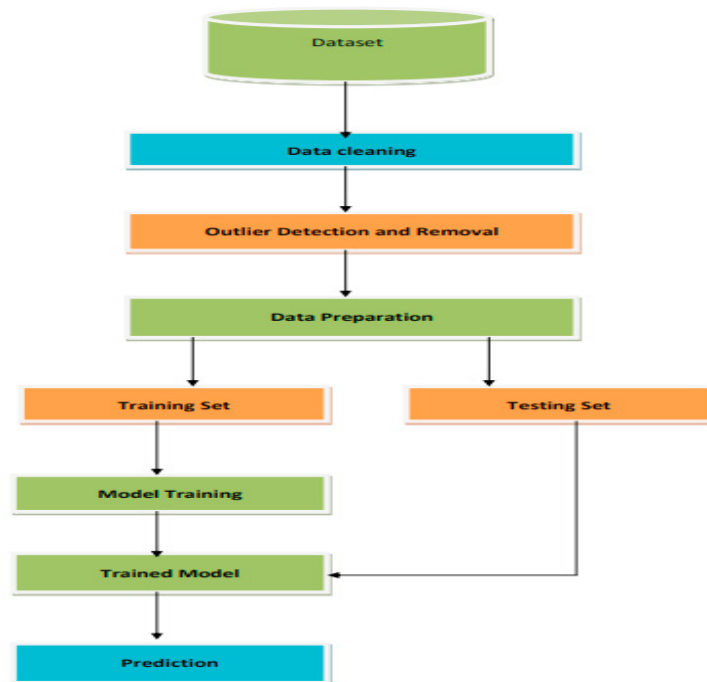


Fig. 8. Proposed Predictive model of power demand [27]

Table 1. Shows the samples of the data set

S.No.	Cid	Date	Pname	Reading	Ckey	Rcount
1	10014678	2013-08-19, 14:48:40	Microwave	0	281420	1
2	10014678	2013-08-19, 14:48:40	TV	0	281420	1
3	10014678	2013-08-19, 14:48:41	Dishwasher	0	281420	1
4	10014678	2013-08-19, 14:48:41	Kettle	0	281420	1
5	10014678	2013-08-19, 14:48:41	Washing machine	.002	281420	1

Table 2. shows the dataset after performing basic pre-processing of data

S.No.	Date	Cid	Reading	Rcount
1	2013-08-19, 14:48:40	10014678	0	1
2	2013-08-19, 14:48:40	10014678	0	1
3	2013-08-19, 14:48:41	10014678	0	1
4	2013-08-19, 14:48:41	10014678	0	1
5	2013-08-19, 14:48:41	10014678	.002	1

4. Results and discussion

The performance of the models for prediction of the meter reading is demonstrated in this section. The performance of the model is measured in terms of MSE which is the loss function for the predictive models. Figure 9 and Table 3 demonstrate the loss of the models during the training and for the validation set the loss is demonstrated in figure 10 and Table 4. According to the obtained performance model 3 demonstrates much better results as compared to other models. The model 3 is a CNN architecture which is trained on the same dataset as the other models has been trained. The comparison of predictive model with existing work and proposed model in this paper in terms of prediction accuracy and time consumption have been shown in table 5 for both the algorithms LSTM and CNN with comparison of SVM and LR.

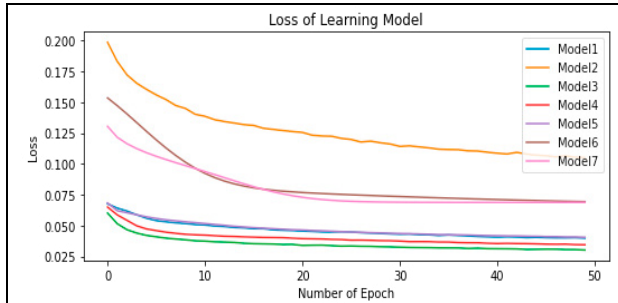


Fig. 9. Loss of the training model

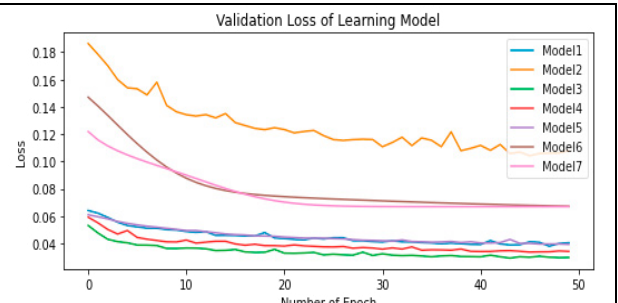


Fig. 10. Loss of validation model

Table 3. Shows the loss of training model.

Epoch Size	Loss function of training model (in MSE)						
	Model-1	Model-2	Model-3	Model-4	Model-5	Model-6	Model-7
0	0.075	0.200	0.065	0.070	0.075	0.165	0.135
10	0.060	0.145	0.040	0.050	0.065	0.090	0.100
20	0.055	0.135	0.035	0.050	0.055	0.085	0.085
30	0.055	0.130	0.030	0.045	0.050	0.080	0.080
40	0.050	0.125	0.027	0.040	0.050	0.080	0.070
50	0.050	0.115	0.025	0.040	0.050	0.070	0.065

Table 4. Shows the loss of validation model

Epoch Size	Loss function of validation model (in MSE)						
	Model-1	Model-2	Model-3	Model-4	Model-5	Model-6	Model-7
0	0.07	0.40	0.05	0.065	0.067	0.16	0.13
10	0.06	0.15	0.04	0.05	0.06	0.09	0.11
20	0.05	0.13	0.03	0.045	0.03	0.085	0.08
30	0.045	0.13	0.025	0.045	0.05	0.09	0.07
40	0.04	0.12	0.02	0.04	0.04	0.09	0.06
50	0.04	0.11	0.01	0.04	0.05	0.09	0.08

Table 3 and table 4 shows that loss function for training model and validation is very low due to increasing epoch size of simulation model during the experimental data taken into consideration.

Table 5. Comparison of predictive model with existing work and proposed model with LSTM and CNN

S.No	Parameters	SVM	LR	LSTM	CNN
1	Accuracy	74.1%	65.27%	73.45%	83.41%
2	Time in sec.	112.81 Sec	84.77	105.71 Sec	69.12

In addition, the performance of prediction by comparing the predicted data and original data is also demonstrated. Figure 11 contains the actual data which is used for testing and the obtained predictions based on the different implemented models are given in this figure.

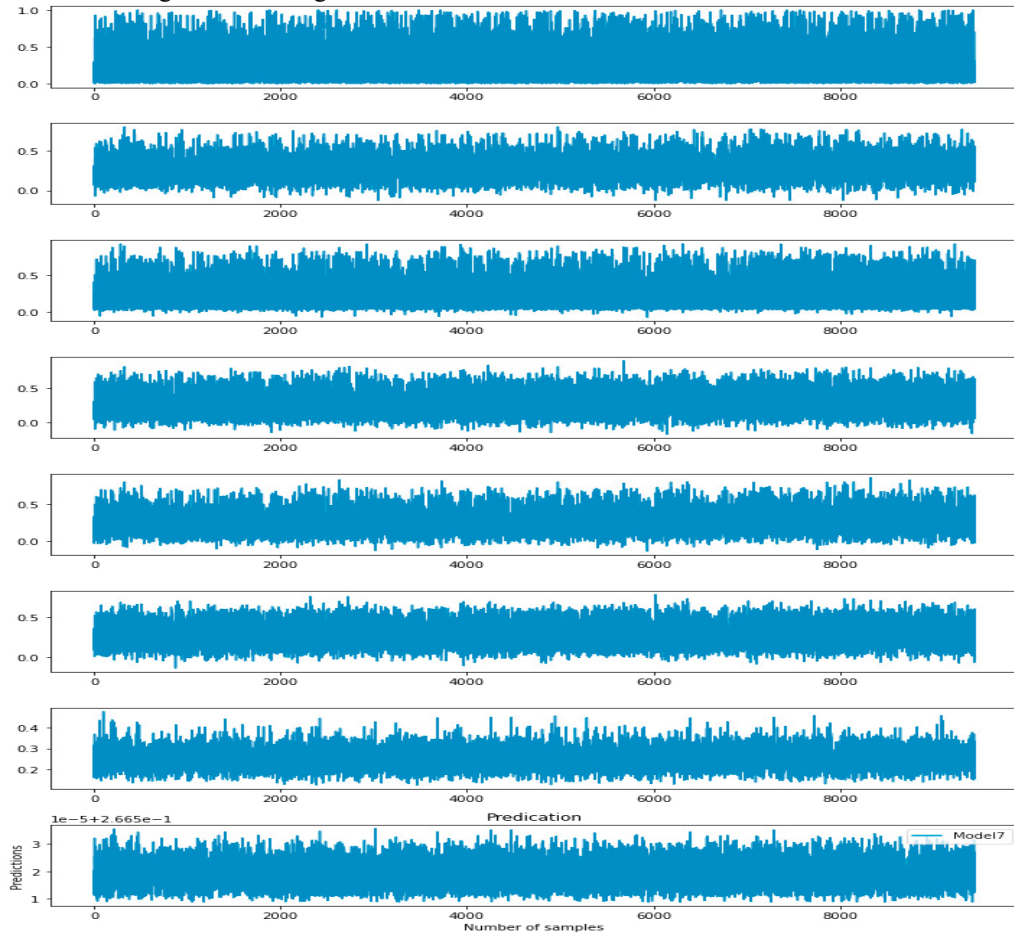


Fig. 11. Predictions by the implemented models in comparison with the original data

5. Conclusion

In this paper, we have concentrated on the implementation of the proposed model for understanding the prediction of power demand and supply. For this we have implemented a client-based machine learning model with two ML methods, LSTM and CNN. The performance of the model for prediction has been obtained and it has been measured in terms of the MSE, which is the loss function of the predictive models. According to the obtained performance, model 3 demonstrates much better results as compared to other models. The model 3 shows prediction accuracy above 83% with quick response time. The model 3 is a CNN architecture which is trained on the same dataset as the other models has been trained. This power distribution technique provides the supply of power according to the need of different sectors. Therefore, in future a simulation is proposed to design and implement which is accurate and real time predicts the load. Additionally, based on this proactive technique, we have tried to reduce the power wastage for better utilization of energy.

Smart meters are crucial for the smooth operation and control of the future smart grid given the situation of increasing electricity demand. In order to deal with the future energy market, load forecasting should be simpler and more accurate with such a capability. In addition, smart meters will be able to send complete and verifiable datasets to a server for immediate analysis.

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